

Student Name:

2005
TRIAL HIGHER SCHOOL CERTIFICATE

MATHEMATICS
Extension 1



General Instructions

Reading Time: 5 minutes
Working Time: 2 hours

- Attempt all questions
- Start each question on a new page
- Each question is of equal value
- Show all necessary working.
- Marks may be deducted for careless work or incomplete solutions
- Standard integrals are printed on the last page
- Board-approved calculators may be used

Question One

Marks

a) Find the remainder when $P(x) = x^3 + 2x - 5$ is divided by $x - 1$.

1

b) Use the table of standard integrals to find $\int \frac{1}{\sqrt{x^2 + 4}} dx$

1

c) Solve the inequality $\frac{2}{x-5} > 1$

2

d) For the points $A(-3, 8)$ and $B(5, -6)$, find the coordinates of the point P which divides the interval AB internally in the ratio 1:3.

2

e) Use the substitution $u = x - 1$ to evaluate $\int_2^4 \frac{x}{(x-1)^2} dx$.

3

f) Solve the equation $\sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$.

3

Question Two (Start a new page)

Marks

a) Find $\frac{d}{dx}(x^3 \tan^{-1} 2x)$

2

b) The function $f(x) = \ln x + 5x$ has a zero near $x = 0.2$

2

Use one application of Newton's Method to find a better approximation, giving your answer correct to 2 decimal places.

c) For the function $f(x) = \cos^{-1}(3x)$

i) Find $f(-\frac{1}{6})$, expressing the answer in radians in exact form.

1

ii) State the domain and range of this function.

1

iii) Neatly sketch $y = f(x)$.

2

d) i) Show that $\frac{x+1}{x+3} = 1 - \frac{2}{x+3}$

1

ii) You are now given that $y = \frac{x+1}{x+3}$, find the equation of the vertical asymptote.

1

iii) Without the use of calculus neatly sketch the graph of $y = \frac{x+1}{x+3}$ showing all the main features.

2

Question Three (Start a new page)

Marks

a) Given that $\int_0^2 f(t) dt = 5$, evaluate $\int_0^1 f(t) dt + \int_1^2 f(t) - 1 dt$.

2

b) If the line $y = mx + b$ is 2 units from the origin, prove that $m^2 + 1 = \frac{b^2}{4}$.

2

c) Prove, by Mathematical Induction, that for all integers $n \geq 1$

4

$$1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1).$$

d) The population N , of Keystown first reached 25 000 on 1 January 2000. The population of Keystown is set to increase according to the equation

$$\frac{dN}{dt} = k(N - 8000)$$

where t represents time in years after the population first reached 25 000. On 1 January 2005, the population of Keystown was 29 250.

i) Verify that $N = 8000 + Ae^{kt}$ is a solution to the above equation where A is a constant.

1

ii) Find the value of A and the value of k correct to 4 decimal places.

2

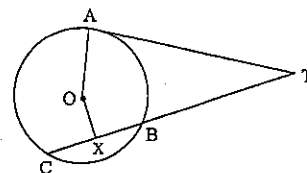
iii) In which year will the population of Keystown reach 50 000?

1

Question Four (Start a new page)

Marks

a)



A, B and C are 3 points on a circle centre O. The tangent at A meets CB produced at T. X is the midpoint of BC.

Neatly copy the diagram onto your answer sheet.

i) Without adding any constructions to the diagram prove that AOXT is a cyclic quadrilateral.

3

ii) Hence state why $\angle AOT = \angle AXT$

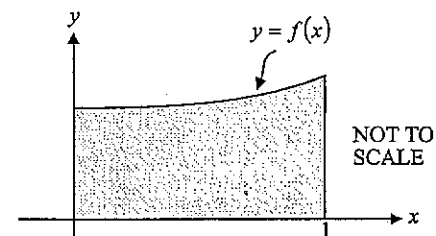
1

b) i) Find the natural domain of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$.

1

ii) The sketch below shows part of the graph of $y = f(x)$. The area under the curve for $0 \leq x \leq 1$ is shaded. Find the area of the shaded region.

2



c) If α, β, γ are the roots of $2x^3 + x^2 - x - 2 = 0$, find the value of:

i) $\alpha + \beta + \gamma$

1

ii) $\alpha\beta\gamma$

1

iii) $\alpha\beta + \alpha\gamma + \beta\gamma$

1

iv) Hence or otherwise, find the value of $(\alpha-1)(\beta-1)(\gamma-1)$.

2

Question Five (Start a new page)

Marks

- a) Air is being pumped into a spherical balloon at the rate of $450 \text{ cm}^3/\text{s}$.

3

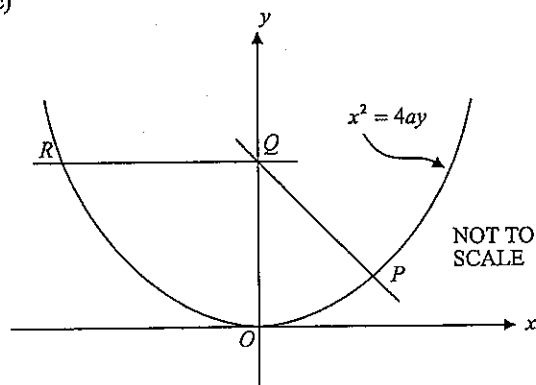
Calculate in exact form the rate at which the radius of the balloon is increasing at the instant when the radius reaches 15cm. $\left[V = \frac{4}{3}\pi r^3 \right]$

- b) The area between the curve $y = \sin x$, the x -axis, the lines $x = 0$ and $x = \frac{3\pi}{4}$ is rotated about the x -axis. Find the volume of the solid formed. Express

3

your answer in exact form.

c)



The diagram above shows the graph of the parabola $x^2 = 4ay$.

The normal to the parabola at the variable point $P(2at, at^2)$, $t > 0$, cuts the y -axis at Q . The point R lies on the parabola in the 2nd quadrant.

- Show that the equation of the normal to the parabola at P is $x + ty = at^3 + 2at$.
- Find the coordinates of R given that QR is parallel to the x -axis.
- Let M be the midpoint of RQ . Find the Cartesian equation of the locus of M .

2

2

2

Question Six(Start a new page)

Marks

- a) A particle moves in a straight line with an acceleration given by

$$\frac{d^2x}{dt^2} = 9(x-2)$$

where x is the displacement in metres from an origin O after t seconds.

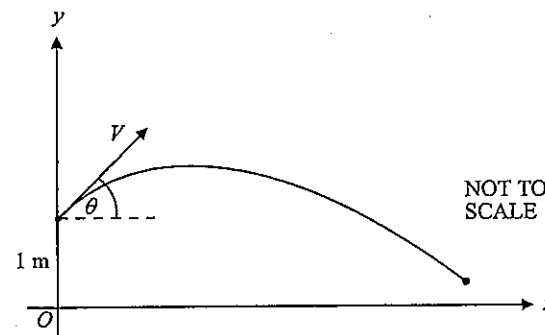
Initially, the particle is 4 metres to the right of O with a velocity $v = -6$.

- Show that $v^2 = 9(x-2)^2$.
- Find an expression for v and hence find x as a function of t .

3

3

- b) A boy throws a ball and projects it with a speed of $V \text{ m s}^{-1}$ from a point 1 metre above the ground. The ball lands on top of a flowerpot in a neighbour's yard.



The angle of projection is θ as indicated in the diagram. The equations of motion of the ball are $\ddot{x} = 0$ and $\ddot{y} = -10$.

It has been found that $y = Vt \sin \theta - 5t^2 + 1$.

- Show that $x = Vt \cos \theta$.
- When the ball is at its maximum height above the ground, it passes directly above a 1.5 metre high fence and clears the fence by 0.5 metres. Show that $V = \frac{\sqrt{20}}{\sin \theta}$.
- Find the value of V given that $\theta = \tan^{-1} \frac{9}{40}$. Give your answer in m/s , correct to 2 decimal places.

1

3

2

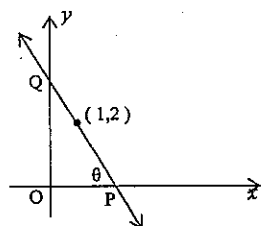
Question Seven(Start a new page)

Marks

a) Find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2}$

2

b)



A line passes through the point $(1, 2)$ and meets the x and y axes at P and Q respectively as shown in the above diagram. $\angle OPQ = \theta$.

i) Show that the equation of the line PQ is given by $y = \tan \theta - x \tan \theta + 2$.

2

ii) Show that the area (A) of $\triangle OPQ$ is given by

$$A = \frac{\tan \theta}{2} + 2 + \frac{2}{\tan \theta}.$$

3

iii) Prove that the area is a minimum when $\tan \theta = 2$.

4

iv) Hence, find the minimum area.

1

END OF EXAM

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

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Question 1.

a) $P(1) = -2$

b) $\ln(x + \sqrt{x^2 + 4}) + C$

c) $2(x-5) > (x-5)^2$

$$2(x-5) - (x-5)^2 > 0$$

$$(x-5)(7-x) > 0 \Rightarrow 5 < x < 7$$

d) $x = \frac{3(-3)+1(5)}{1+3}, y = \frac{3(8)+1(-6)}{1+3}$

$\therefore P$ has coordinates $\left(-1, \frac{9}{2}\right)$.

e) Let $u = x-1 \Rightarrow x = u+1$ and $du = dx$.
Also when $x=4, u=3$ and when $x=2, u=1$.

$$\therefore I = \int_1^3 \frac{u+1}{u^2} du \Rightarrow I = \int_1^3 \frac{1}{u} + u^{-2} du$$

$$\therefore I = \left[\ln u - \frac{1}{u} \right]_1^3 = \left(\ln 3 - \frac{1}{3} \right) - (\ln 1 - 1)$$

$$= \frac{2}{3} + \ln 3.$$

f) $2 \sin \theta \cos \theta - \cos \theta = 0$

$$\cos \theta (2 \sin \theta - 1) = 0$$

$$\therefore \cos \theta = 0 \text{ or } \sin \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2} \text{ and } \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Question 2.

a)

$$\frac{d}{dx}(x^3 \tan^{-1} x) = \tan^{-1} 2x \times 3x^2 + x^3 \times \frac{1}{1+(2x)^2} \times 2$$

$$= 3x^2 \tan^{-1} 2x + \frac{2x^3}{1+4x^2}$$

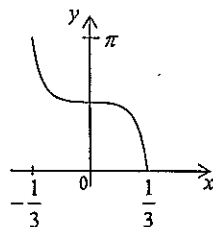
b) $f(x) = \ln x + 5x \Rightarrow f'(x) = \frac{1}{x} + 5$

$$\therefore x = 0.2 - \frac{\ln(0.2)+1}{10} \Rightarrow x = 0.26$$

c) i) $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

ii) D: $-\frac{1}{3} \leq x \leq \frac{1}{3}$, R: $0 \leq y \leq \pi$

iii)

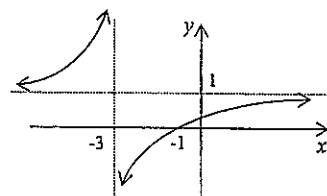


d) $LHS = \frac{x+3-2}{x+3} = \frac{x+3}{x+3} - \frac{2}{x+3}$

$$= 1 - \frac{2}{x+3} = RHS.$$

ii) $x = -3$.

iii)



Question 3.

a) $I = \int_0^1 f(t) dt + \int_1^2 f(t) dt - \int_1^2 1 dt$

$$= \int_0^2 f(t) dt - [t]_1^2 = 5 - [2-1] = 4$$

b) $mx - y + b = 0$ and using the perpendicular distance formula with (x_1, y_1) as $(0, 0)$ and $a = m, b = -1, c = b, d = 2$ we obtain:

$$2 = \frac{|m(0) + (-1)(0) + b|}{\sqrt{m^2 + 1}} \Rightarrow 2 = \frac{b}{\sqrt{m^2 + 1}}$$

$$\therefore 4 = \frac{b^2}{m^2 + 1} \Rightarrow m^2 + 1 = \frac{b^2}{4}$$

c) When $n=1$, $LHS=1$, $RHS = \frac{1}{6}(1)(3)(2) = 1$

$\therefore$ true for $n=1$.

Assume true for $n=k$

$$\text{i.e. } S_k = \frac{1}{6}k(4k-1)(k+1)$$

Prove true for $n=k+1$

i.e. Prove that $S_{k+1} = S_k + T_{k+1}$

$$RHS = \frac{1}{6}k(4k-1)(k+1) + (k+1)(2k+1)$$

$$= \frac{k(4k-1)(k+1) + 6(k+1)(2k+1)}{6}$$

$$= \frac{(k+1)[k(4k-1) + 6(2k+1)]}{6}$$

$$= \frac{(k+1)(4k^2 + 11k + 6)}{6}$$

$$= \frac{1}{6}(k+1)(k+2)(4k+3) = S_{k+1}.$$

Thus proved true for $n=1$, assumed true for $n=k$ and proved true for $n=k+1$, so by Mathematical Induction it is true for $n=1+1=2$, $n=2+1=3$, and so on for all positive integers n .

d) i) $N = 8000 + Ae^{kt} \Rightarrow Ae^{kt} = N - 8000$

$$\therefore \frac{dN}{dt} = kAe^{kt} = k(N - 8000)$$

ii) When $t=0, N=25000$

$$\therefore 25000 = 8000 + Ae^0 \Rightarrow A = 17000$$

When $t=5, N=29250$

$$\therefore 29250 = 8000 + 17000e^{5k}$$

$$e^{5k} = \frac{21250}{17000}$$

$$5k = \ln\left(\frac{21250}{17000}\right) \Rightarrow k = 0.0446$$

iii) $50000 = 8000 + 17000e^{0.0446t}$

$$e^{0.0446t} = \frac{42000}{17000} \Rightarrow t = 20$$

$\therefore$ In 2020 the population will reach 50 000.

Question 4.

a) i) The tangent AT meets radius OA at 90° .
 $\therefore \angle OAT = 90^\circ$.

Also, a line from the centre that bisects a chord is perpendicular to the chord

$$\therefore \angle OXT = 90^\circ$$

Hence $AOXT$ is a cyclic quad (opp. $\angle$'s supp.)

ii) Since $AOXT$ is cyclic $\angle AOT = \angle AXT$
[Angles at circumference standing on same arc AT].

b) i) $4 - x^2 > 0 \Rightarrow -2 < x < 2$.

ii) $A = \int_0^1 \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_0^1$

$$= \sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0) = \frac{\pi}{6} u^2$$

Question 4(continued).

c) i) $\alpha + \beta + \gamma = -\frac{b}{a} = -\frac{1}{2}$

ii) $\alpha\beta\gamma = -\frac{d}{a} = 1$

iii) $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -\frac{1}{2}$

iv) On Expansion

$$(\alpha-1)(\beta-1)(\gamma-1) = \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1$$

$$= 1 - \left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right) - 1 = 0$$

Question 5.

a) $\frac{dV}{dt} = 450$, $V = \frac{4}{3}\pi r^3$, $\frac{dV}{dr} = 4\pi r^2$, $r = 15 \text{ cm}$

$\therefore$ using the chain rule $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

$$\therefore 450 = 4\pi r^2 \cdot \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{450}{4\pi \times 15^2} = \frac{1}{2\pi} \text{ cm/s.}$$

b) $V = \pi \int_0^{\frac{3\pi}{4}} \sin^2 x \, dx$ $\left[\cos 2\theta = 1 - 2\sin^2 \theta \right]$

$$\therefore \left[\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) \right]$$

$$\therefore V = \pi \int_0^{\frac{3\pi}{4}} \frac{1}{2}(1 - \cos 2x) \, dx = \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{3\pi}{4}}$$

$$= \frac{\pi}{2} \left[\left(\frac{3\pi}{4} - \frac{1}{2} \sin \frac{3\pi}{2} \right) - (0 - 0) \right]$$

$$= \frac{\pi}{2} \left[\frac{3\pi}{4} + \frac{1}{2} \right] = \frac{3\pi^2 + 2\pi}{8} u^3$$

c) i) $y = \frac{x^2}{4a} \Rightarrow \frac{dy}{dx} = \frac{x}{2a} \Rightarrow m_T = t$ when $x = 2at$

$\therefore m_N = -\frac{1}{t} \Rightarrow \text{Eq}^n$ of Normal: $y - at^2 = -\frac{1}{t}(x - 2at)$

$\therefore ty - at^3 = 2at - x \Rightarrow x + ty = 2at + at^3$

ii) Using (i) and putting $x = 0 \Rightarrow y = 2a + at^2$

$\therefore Q$ has coordinates $(0, 2a + at^2)$

and since R lies on the parabola then when $y = 2a + at^2$, $x^2 = 4a(2a + at^2) = 4a^2(2 + t^2)$

which gives $x = -2a\sqrt{2 + t^2}$ since R is in the second quadrant.

$\therefore R$ has coordinates $(-2a\sqrt{2 + t^2}, 2a + at^2)$.

iii) M has coordinates $(-a\sqrt{2 + t^2}, 2a + at^2)$

For the locus of M : $x = -a\sqrt{2 + t^2}$

$\therefore x^2 = a^2(2 + t^2) = a(2a + at^2)$

$= ay$

Hence $x = -\sqrt{ay}$ since R is in the 2nd Quad. is the locus of M .

Question 6.

a) i) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 9x - 18$

$\therefore \frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + C$

$\frac{1}{2}(-6)^2 = \frac{9(4)^2}{2} - 18(4) + C \Rightarrow C = 18$

$\therefore \frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + 18$

$v^2 = 9x^2 - 36x + 36$

$= 9(x^2 - 4x + 4) = 9(x - 2)^2$

ii) $v = -3(x - 2)$ since $v < 0$ when $t = 0$

$\frac{dx}{dt} = -3(x - 2) \Rightarrow \frac{dt}{dx} = \frac{-1}{3(x - 2)}$

Question 6 a) ii) continued.

$t = -\frac{1}{3} \ln(x - 2) + C$

when $t = 0, x = 4 \Rightarrow 0 = -\frac{1}{3} \ln 2 + C \Rightarrow C = \frac{1}{3} \ln 2$

$\therefore t = \frac{1}{3} [\ln 2 - \ln(x - 2)] \Rightarrow 3t = \ln \left(\frac{2}{x - 2} \right)$

$\therefore e^{3t} = \frac{2}{x - 2} \Rightarrow x - 2 = \frac{2}{e^{3t}}$

$\therefore x = 2 + 2e^{-3t} \Rightarrow x = 2(1 + e^{-3t})$

b) i) When $t = 0, x = 0, \dot{x} = V \cos \theta$ (a constant)

$\ddot{x} = 0 \Rightarrow \dot{x} = C \Rightarrow \dot{x} = V \cos \theta$

$\therefore x = Vt \cos \theta + C$ but $C = 0$ by I.C.

Hence $x = Vt \cos \theta$.

ii) $y = Vt \sin \theta - 5t^2 + 1 \Rightarrow \dot{y} = V \sin \theta - 10t$

At maximum height $\dot{y} = 0 \Rightarrow t = \frac{V \sin \theta}{10}$

and when $t = \frac{V \sin \theta}{10}, y = 2$

$\therefore 2 = V \left(\frac{V \sin \theta}{10} \right) \sin \theta - 5 \left(\frac{V \sin \theta}{10} \right)^2 + 1$

$2 = \frac{V^2 \sin^2 \theta}{10} - \frac{5V^2 \sin^2 \theta}{100} + 1$

$200 = 10V^2 \sin^2 \theta - 5V^2 \sin^2 \theta + 100$

$\therefore V^2 \sin^2 \theta = 20 \Rightarrow V^2 = \frac{20}{\sin^2 \theta} \Rightarrow V = \frac{\sqrt{20}}{\sin \theta}$

iii) $\tan \theta = \frac{9}{40} \Rightarrow \sin \theta = \frac{9}{41}$ using Pythagoras

$\therefore V = \frac{\sqrt{20}}{\frac{9}{41}} = \frac{41\sqrt{20}}{9}$

$\therefore V \approx 20.37 \text{ ms}^{-1}$ correct to 2 decimal places.

Question 7.

a) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{3x^2}$

$$= \frac{2}{3} \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\sin x}{x}$$

$$= \frac{2}{3} \cdot 1 \cdot 1 = \frac{2}{3}$$

b) i) Gradient of $PQ = \tan(180 - \theta) = -\tan \theta$

$\therefore y - 2 = -\tan \theta (x - 1)$

$y - 2 = -x \tan \theta + \tan \theta$

$y = \tan \theta - x \tan \theta + 2$

ii) Using i) when $x = 0, y = 2 + \tan \theta$

and when $y = 0, x = \frac{2 + \tan \theta}{\tan \theta}$

$\therefore \text{Area } \Delta OPQ = \frac{1}{2} OP \times OQ$

$$= \frac{1}{2} \left(\frac{2 + \tan \theta}{\tan \theta} \right) (2 + \tan \theta)$$

$$= \frac{1}{2} \left(\frac{4 + 4 \tan \theta + \tan^2 \theta}{\tan \theta} \right)$$

$$= \frac{1}{2} \left(\frac{4}{\tan \theta} + 4 + \frac{\tan^2 \theta}{\tan \theta} \right)$$

$$= \frac{2}{\tan \theta} + 2 + \frac{\tan \theta}{2}$$

iii) In part ii) let $t = \tan \theta$

$\therefore A = \frac{t}{2} + 2 + \frac{2}{t} = \frac{t}{2} + 2 + 2t^{-1}$

For a minimum area $\frac{dA}{dt} = 0$ and $\frac{d^2A}{dt^2} > 0$

$\frac{dA}{dt} = \frac{1}{2} - 2t^{-2} = \frac{1}{2} - \frac{2}{t^2}$ and $\frac{d^2A}{dt^2} = \frac{4}{t^3}$

$\therefore \frac{2}{t^2} = \frac{1}{2} \Rightarrow t^2 = 4 \Rightarrow t = 2$, also $\frac{d^2A}{dt^2} = \frac{1}{2}$

$\therefore$ minimum area when $\tan \theta = 2$

iv) $\therefore \text{Minimum Area} = 1 + 2 + 1 = 4 u^2$.